Reneal eLearning Lab Operations

Report 1: General System & Lab Usage Summary

School Name: Amani Abeid Karume Secondary School

Overall Status: EXCELLENT (Rank 1)

Academic Cycle: 2024 Academic Year

Active Terminals: 42 Thin Clients

42	100%	Rank 1
ACTIVE THIN CLIENTS	PXE NETWORK BOOT HEALTH	SYSTEM PERFORMANCE

1. Executive Summary

Amani Abeid Karume Secondary School currently maintains the top overall performance ranking (Rank 1) within the Reneal Lab Statistics Tracker for the 2024 academic cycle. The computer lab operates with high hardware availability, featuring 42 fully functional thin-client terminals connected via Linux Terminal Server Project (LTSP) PXE network booting. The infrastructure demonstrates exceptional uptime, consistent student session engagement, and efficient local network resource delivery.

2. Hardware & Infrastructure Status

The computer laboratory infrastructure consists of a high-capacity Ubuntu Linux central host server powering 42 thin client workstations. All terminal boot sequences execute over a gigabit local network using PXE protocol without disk image corruption or boot failures.

Component	Specification / Quantity	Operational Status
Central Host Server	Ubuntu Linux (LTSP configured)	Operational / Optimal Uptime
Thin Client Terminals	42 Active Stations	100% Fully Functional
Local Network Booting	PXE / TFTP Image Server	Zero Boot Errors Reported
Power Protection	Eaton Line-Interactive UPS	Stable / Battery Healthy

3. General Usage & eLearning Engagement

Students and teachers actively utilize the digital lab for daily academic coursework and self-guided research. Key educational assets including Wikipedia Offline, Khan Academy Lite, and local coding repositories witness continuous daily access. The facility also serves as a primary hub for web development learning sessions utilizing HTML, CSS, and basic programming environments.

4. Key Operational Recommendations

To sustain Rank 1 performance, routine weekly checks on host server log files and physical dust clearing from network switches and UPS battery ventilation fans are recommended during scheduled school breaks.